

CS102: Relational Database Management Systems & SQL

Chapter 1: Relational Model & SQL

A Database Management System (DBMS) is software designed to define, manipulate, retrieve, and manage structured data.

The relational model represents data as tables consisting of rows (tuples) and columns (attributes).

Primary key uniquely identifies each record in a relation. Foreign key establishes referential integrity between tables.

Relational algebra operations include Selection (σ), Projection (π), Union, Set Difference, Cartesian Product, and Natural Join.

Data Definition Language (DDL) commands include CREATE, ALTER, DROP, and TRUNCATE.

Data Manipulation Language (DML) commands include SELECT, INSERT, UPDATE, and DELETE.

Chapter 2: Relational Database Normalization

Database normalization minimizes data redundancy and eliminates insertion, update, and deletion anomalies.

Functional Dependency ($X \rightarrow Y$) means X uniquely determines Y.

First Normal Form (1NF): Requires atomic attribute values with no repeating groups.

Second Normal Form (2NF): Requires 1NF and no partial dependencies where non-prime attributes depend on a subset of a composite primary key.

Third Normal Form (3NF): Requires 2NF and no transitive dependencies where non-prime attributes depend on other non-prime attributes.

Boyce-Codd Normal Form (BCNF): A stricter version of 3NF where for every functional dependency $X \rightarrow Y$, X must be a super key.

Chapter 3: Transactions & ACID Properties

A transaction is a logical unit of database processing executed as a single unit.

ACID properties guarantee reliability in database transactions:

Atomicity: All operations in a transaction succeed, or the entire transaction is rolled back.

Consistency: Database transitions from one valid state to another valid state preserving integrity constraints.

Isolation: Concurrent transactions execute independently without interfering with each other.

Durability: Committed transactions persist permanently even during system failures.

Concurrency control protocols such as Two-Phase Locking (2PL) and Timestamp Ordering ensure serializability.

Deadlocks in DBMS occur when transactions wait indefinitely for locks held by each other.

Chapter 4: Indexing & Query Optimization

Database indexing improves data retrieval speed at the cost of additional storage and write overhead.

B-Trees and B+ Trees are balanced search trees widely used for disk-based indexing because they maintain sorted data and allow efficient search, insertion, and deletion in $O(\log N)$ time.

Hash indexes provide $O(1)$ average time complexity for exact match equality queries.

Query optimizers evaluate query execution plans and choose optimal join algorithms such as Nested Loop Join, Hash Join, and Sort-Merge Join based on cost estimation.